Multimodal Ground-Based Remote Sensing Cloud Classification via Learning Heterogeneous Deep Features

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***Abstract*— Recently, multimodal cloud samples are utilized to learn completed feature representations for cloud classification. However, the existing methods neglect the related information from other multimodal cloud samples in the learning process, which leads to inadequate learning. In this article, we propose a novel deep model to learn heterogeneous deep features (HDFs) for multimodal ground-based remote sensing cloud classification. Specifically, we first design the convolutional neural network (CNN) extractor to combine the visual information and the multimodal information (MI) to obtain the CNN-based features of multimodal cloud samples. Afterward, we treat the CNN-based features of multimodal cloud samples as the nodes of graph, and utilize the similarity between nodes as the adjacency matrix. We feed the graph and the adjacency matrix into the graph convolutional network (GCN) extractor to obtain the GCN-based features that could capture correlations among multimodal cloud samples using graph convolutional layers. After obtaining CNN-based features and GCN-based features, we concatenate the two kinds of heterogeneous features to represent the multimodal cloud samples. As a result, the concatenated feature contains the visual information, the MI and the related information among multimodal cloud samples. We conduct a series of experiments on the multimodal ground-based cloud database (MGCD), and the experimental results verify that the proposed HDF outperforms state-of-the-art methods.**

***Index Terms*— Convolutional neural network (CNN), graph convolutional network (GCN), heterogeneous features, multimodal ground-based remote sensing cloud classification.**

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I. Introduction

CLOUDS are visible aggregations of a large number of tiny water droplets or ice crystals suspended in the air. They have a great influence on the hydrological cycle and the energy balance on both local and global scales due to evolutions of volume, thickness, and shape of clouds [1]–[3]. Consequently, many researchers focus on observations of volume, thickness, and shape of clouds [4]–[7]. Since clouds are changeable and their boundaries ambiguous, cloud classification is significantly challenging. At present, there are mainly two observation methods for cloud classification, that is, satellite remote sensing [8] and ground-based remote sensing [3]. The satellite remote sensing observation has a huge observation range, and collects global cloud data. Hence, this kind of cloud observation is appropriate for large-scale climate research. On the contrary, the ground-based remote sensing observation captures local clouds from ground. Correspondingly, ground-based remote sensing cloud images contain more local distribution information, and therefore they are very useful for classifying various cloud types.

An amount of automatic classification methods for ground-based remote sensing cloud images are proposed [9]–[11], [14], [15]. Among them, some methods [1], [2], [16]–[19] extract hand-crafted features to represent texture, shape, and color information of ground-based cloud images. However, these features are predefined and not adaptive to different sample distributions. Recently, convolutional neural network (CNN) has been successfully applied in many research fields [20]–[24], and it also shows the superiority in the ground-based remote sensing cloud classification task [13]–[15], [25]. Compared to the traditional methods, CNN-based methods are data-dependent, which could learn features according to data distributions. The above automatic classification methods only consider the visual information of ground-based remote sensing clouds. As we know, temperature and air pressure influence the formation of clouds, and wind speed influences the shape of clouds [26]. Hence, cloud types are influenced by many factors, and it cannot represent clouds completely using only the visual information. Various approaches [26]–[29] are proposed to integrate the multimodal information (MI) and the visual information to obtain completed representations of ground-based remote

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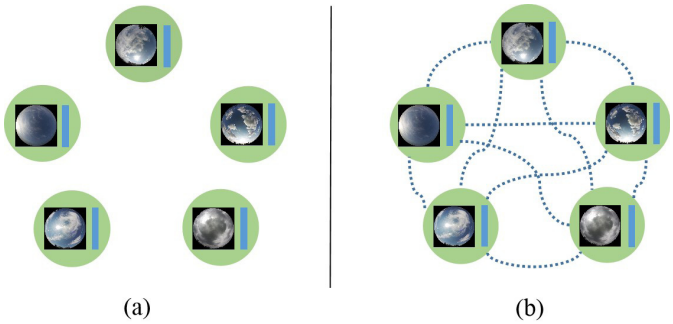


Fig. 1. Multimodal cloud samples in (a) are independent in the learning process, and the multimodal cloud samples in (b) are connected by edges to reflect correlations among samples. Each circle represents a multimodal cloud sample including a cloud image and the MI.

sensing clouds. The MI includes temperature, pressure, humidity, and wind speed. These approaches are known as multimodal ground-based remote sensing cloud classification.

However, existing multimodal ground-based remote sensing cloud classification approaches consider only a single multimodal cloud sample in the learning process, which neglects the related information from other multimodal cloud samples. As shown in Fig. 1(a), the multimodal cloud samples are mutually independent in the existing approaches. Without these connections, the multimodal cloud samples cannot learn the information from other samples.

In this article, we propose a novel deep model to learn heterogeneous deep features (HDFs) for multimodal ground-based remote sensing cloud classification. We propose the CNN extractor and the graph convolutional network (GCN) extractor to learn CNN-based features and GCN-based features of multimodal cloud samples. The CNN extractor is utilized to combine the visual information and the MI. The GCN extractor could capture correlations among multimodal cloud samples through constructing graph structure data as shown in Fig. 1(b). The CNN-based features and the GCN-based features are concatenated to obtain the final representations of multimodal cloud samples.

The contributions of the proposed HDF are summarized in three aspects. First, to the best of our knowledge, we are the first one to utilize GCN for multimodal ground-based remote sensing cloud classification. Second, we fuse two kinds of heterogeneous features, that is, CNN-based features and GCN-based features, to represent multimodal cloud samples, which not only contains the multimodal and visual information but also injects the correlation information among samples into the feature representations. Finally, the proposed HDF is proven to be more effective than state-of-the-art methods on the large-scale multimodal cloud data set.

The remainder of this article is organized as follows. Section II gives a review of related work, Section III describes the details of the proposed HDF, Section IV shows the experimental results to verify the advantages of HDF, and finally, we draw a conclusion in Section V.

II. Related Work

A. Ground-Based Remote Sensing Cloud Classification Using CNN

Some methods [9], [11]–[13], [30] usually employ hand-crafted features for ground-based remote sensing cloud classification. For example, Dev *et al.* [12] proposed an improved texton-based texture classification method which applies a set of rotation-invariant Gabor-like filters to generate image descriptors. Ye *et al.* [13] proposed a practical method to detect clouds and recognize all cloud types simultaneously. They extracted the multiview features of superpixels and utilized the feature transformation method, that is, metric learning (ML) and subspace alignment (SA), to obtain the feature representations of superpixels. Compared to the traditional methods, the ground-based remote sensing cloud classification using CNN has achieved remarkable improvement [14], [15], [25], [31]–[33]. The CNN-based methods mainly include two types, that is, one-step and two-step methods. The one-step method directly utilizes the features extracted from CNN models as the representations of ground-based remote sensing cloud images. For example, Shi *et al.* [25] presented to extract local texture information of ground-based remote sensing cloud images from shallow convolutional layers of CNN. Zhang *et al.* [14] proposed the CloudNet which contains five convolutional layers and two fully connected layers for the cloud classification task, and treated the output of the final convolutional layer as the feature representation of cloud image. The 3-D-CNN [15] was proposed to extract features of cloud images from the convolutional layers so as to learn the texture and temporal information.

The two-step method initially extracts features from CNN models, and then reprocesses them to obtain the final representations. Ye *et al.* [31] first extracted cloud features from CNN, and then encoded them using Fisher vectors (FV) to produce final representations. The salient dual activations aggregation (SDAA) [32] was presented to aggregate features from both shallow and deep convolutional layers using a weighted strategy. Zhang *et al.* [33] proposed the transfer deep local binary patterns (TDLBPs) to extract feature maps from the CNN model and then applied local binary patterns (LBPs) on the feature maps to represent cloud images. Except for feature extraction methods, Li *et al.* [34] proposed the dual-guided loss (DGL) to optimize CNN models, so that the CNN model has a balanced learning ability for both easy-classified and hard-classified cloud images.

B. Multimodal Ground-Based Remote Sensing Cloud Classification Using CNN

Since the MI is closely related to the formation of clouds, many researchers employ the MI for ground-based remote sensing cloud classification [26]–[29]. For instance, Liu and Li [27] concatenated the MI with deep visual features to represent ground-based remote sensing clouds. Liu *et al.* [26] presented the joint fusion CNN (JFCNN) which jointly learns deep visual features and deep multimodal features using the joint fusion layer in a unified framework. In [28], hierarchical multimodal fusion (HMF) was proposed to fuse deep visual features and deep multimodal features in different layers in order to capture more information between these two kinds of features. The deep tensor fusion network (DTFN) [29] was utilized to integrate the visual information and the MI at the tensor level, which is beneficial to maintaining the spatial information of cloud images.

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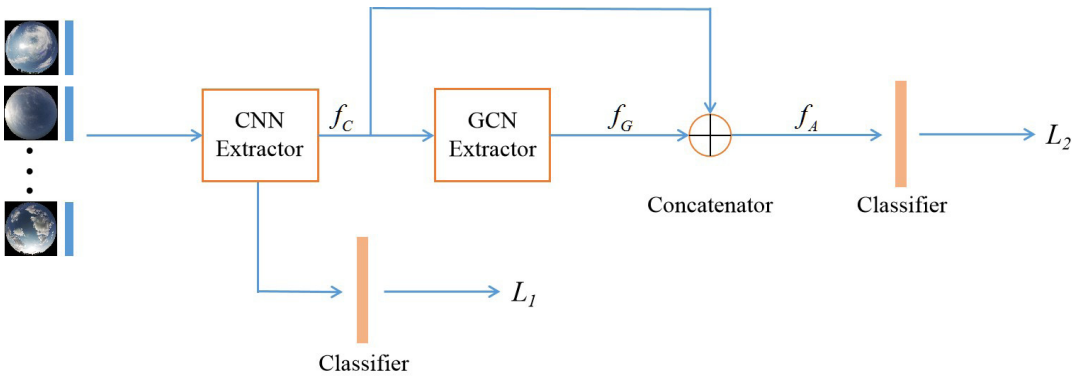


Fig. 2. Framework of the proposed HDF. f_C, f_G, and f_A are CNN-based features, GCN-based features, and concatenated features of multimodal cloud samples. L_1 and L_2 are the cross-entropy loss.

C. Graph Convolutional Networks

GCN is proposed to generalize the convolution operation on the graph structure data [35]. Since most kinds of data can be constructed as a graph, GCN has been applied in many research fields [36]–[40], such as skeleton-based action recognition, web-scale recommender system construction and polypharmacy side effects modeling. The mainstream approaches of GCN include spectral domain methods and spatial domain methods. The spectral domain methods [41]–[43] implement the Fourier transformation and the convolution operation on graph structure data with the spectral graph theory.

The spatial domain methods [36], [44], [45] directly conduct the convolution operation based on receptive fields which contain the central node and its neighbors on the graph. In graph construction, some methods [45], [46] connect nodes with the fixed values, and utilize 0 or 1 to determine whether the two nodes are connected or not. Meanwhile, some other methods [47]–[49] are data-dependent, and assign different weight values between graph nodes to determine the connection strengths. In this article, we apply multimodal cloud samples and their similarities to construct the graph, and employ graph convolutional layers to learn GCN-based features in order to consider the related information among multimodal cloud samples.

III. Approach

In this section, we first present an overview of the proposed HDF, and then describe the major parts of HDF, that is, CNN extractor and GCN extractor in detail. Finally, we introduce how to fuse the heterogeneous features for multimodal cloud sample representations.

A. Overview of HDF

Although the performance of multimodal ground-based remote sensing cloud classification has been improved, the existing methods focus on learning features from single multimodal cloud sample, which neglects the correlations among samples. In this article, we propose HDF to address this drawback. The framework of HDF is shown in Fig. 2.

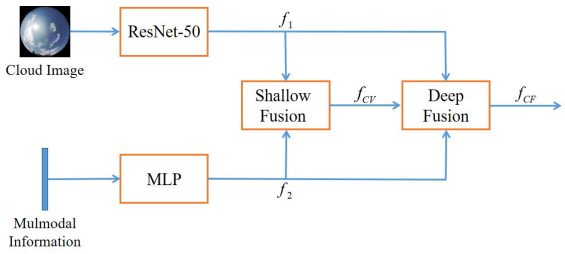


Fig. 3. Framework of CNN extractor. f_1 and f_2 indicate the visual feature and the multimodal feature. f_{CV} and f_{CF} are the CNN-based features of vanilla version and full version, respectively.

We first present the proposed HDF in the following three aspects briefly.

1) CNN Extractor: The target of CNN extractor is to combine the visual information and the multimodal information to obtain CNN-based features of multimodal cloud samples. As shown in Fig. 3, the cloud image and the multimodal information are fed into the CNN extractor, and the output is the CNN-based feature which contains both the visual and multimodal information. In addition, we apply two kinds of CNN extractors, that is, vanilla version and full version. The vanilla version utilizes the shallow fusion strategy while the full version applies the deep fusion strategy.

2) GCN Extractor: The GCN extractor applies GCN to capture correlations among multimodal cloud samples and obtain GCN-based features of multimodal cloud samples. Specifically, we treat CNN-based features of multimodal cloud samples as the nodes and connect the nodes using edges in order to construct a graph where the edges reflect the correlations among the nodes. In order to fully discover the correlations among the nodes, we regard the similarity between the nodes as the edge and utilize all similarities to obtain the adjacency matrix. With the graph, the related information could be learned using the graph convolutional layers in GCN because the information of the nodes is propagated along the graph edges. As a result, the GCN-based features utilize the related information among multimodal cloud samples to promote the discrimination of the representations of multimodal cloud samples.

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TABLE I
STRUCTURE OF RESNET-50

Name	Output Size	Filters	Padding
Conv1	112×112	[7 × 7, 64], stride = 2	(3, 3)
Max pooling	112×112	3 × 3, stride = 2	(1, 1)
Conv2_x	56×56	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} (0,0) \\ (1,1) \\ (0,0) \end{bmatrix} \times 3$
Conv3_x	28×28	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} (0,0) \\ (1,1) \\ (0,0) \end{bmatrix} \times 4$
Conv4_x	14×14	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix} \times 6$	$\begin{bmatrix} (0,0) \\ (1,1) \\ (0,0) \end{bmatrix} \times 6$
Conv5_x	7×7	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} (0,0) \\ (1,1) \\ (0,0) \end{bmatrix} \times 3$
FC	1×1	Average pooling	

3) Feature Fusion: After obtaining the two kinds of heterogeneous features, that is, CNN-based features and GCN-based features, we propose to concatenate them for representing multimodal cloud samples so as to take full advantages of them. Specifically, a concatenator is utilized to concatenate the two kinds of heterogeneous features. Hence, besides the multimodal and visual information, the related information from other multimodal cloud samples is injected into the final representations so as to enhance the discrimination.

B. CNN Extractor

Fig. 3 shows the structure of CNN extractor. First, we employ ResNet-50 [21] and multilayer perceptron (MLP) to learn deep features of multimodal cloud samples. The structure of ResNet-50 is illustrated in Table I. The output of average pooling layer in ResNet-50 is regarded as the visual feature. The MLP includes six FC layers, and the neuron number of FC layers is set to 64, 128, 256, 512, 1024, and 2048, respectively. The output of MLP is the multimodal feature. In Fig. 3, there are two fusion strategies, that is, shallow fusion and deep fusion, where the outputs of shallow fusion and deep fusion of CNN extractor are denoted as f_{CV} and f_{CF}, respectively. Therefore, f_{CV} and f_{CF} are also denoted as the CNN-based features of vanilla version and full version, respectively.

Afterward, we adopt two fusion strategies in CNN extractor, that is, vanilla version [26] and full version [28]. The vanilla version is the shallow fusion of visual features and multimodal features, and it is defined as

$$f_{\mathrm{CV}} = \left[f_1^T, \alpha f_2^T\right]^T \tag{1}$$

where f_1 and f_2 indicate the visual feature and the multimodal feature respectively, $\alpha \in [0, 1]$ is the weight to determine the importance of multimodal feature, and $[\cdot]$ means the concatenation operation. Here, $f_{\mathrm{CV}} \in \mathbb{R}^d$ is the CNN-based feature of vanilla version, where d is the feature dimensionality.

To further fuse visual features and multimodal features, the full version adopts the deep fusion strategy. As shown in Fig. 3, the deep fusion strategy combines the visual feature, the multimodal feature and the CNN-based feature of vanilla version, and it is formulated as

$$f_{\mathrm{CF}} = \left[\lambda_1 f_1^T, f_{\mathrm{CV}}^T, \lambda_2 f_2^T\right]^T \tag{2}$$

where λ_1 and λ_2 are the weights to balance the importance of f_1 and f_2, f_{CF} is the CNN-based feature of full version. We employ the cross-entropy loss to optimize the CNN extractor

$$L_1 = -\sum_{j=1}^{J} q_j \log y_j \tag{3}$$

where J is the number of cloud types, q_j is the real probability of the multimodal cloud sample belonging to the jth cloud type, and $q_j = 1$ when j is the ground-truth label, otherwise, $q_j = 0$. Here, $y_j \in [0, 1]$ indicates the predicted probability of the multimodal cloud sample belongs to the jth cloud type, and it is the output of the classifier which is an FC layer followed by the softmax function. The softmax function is defined as

$$y_j = \frac{e^{x_j}}{\sum_{t=1}^{J} e^{x_t}} \tag{4}$$

where x_j and x_t are the outputs of the jth neuron and the tth neuron in the FC layer respectively, and J is the number of cloud types. In order to consider the numerical instability, we utilize the modified softmax function as follows:

$$y_j = \frac{e^{(x_j - M)}}{\sum_{t=1}^{J} e^{(x_t - M)}}, \quad M = \max_t x_t \tag{5}$$

where M is the maximum output value of FC layer, and J is the number of cloud types. In the modified softmax function, the maximum value of exponential term is 0, which could effectively avoid the problem of quite large exponential term.

C. GCN Extractor

GCN has the ability to propagate the information between the central sample and its neighboring samples in the graph, which could learn the related information from other multimodal cloud samples. There are mainly two steps in the GCN extractor. The first step is to construct a graph and the second step is to extract GCN-based features. The graph in GCN consists of nodes and edges so that the information of each node could propagate along the edges. Specifically, the CNN-based features of multimodal cloud samples are treated as the nodes, and the similarities between nodes are computed to represent their correlations. The more similar the two nodes are, the greater their correlation is. The similarity between the ith node and the jth node is defined as

$$a_{ij} = f_C^i \cdot f_C^{j^T} \tag{6}$$

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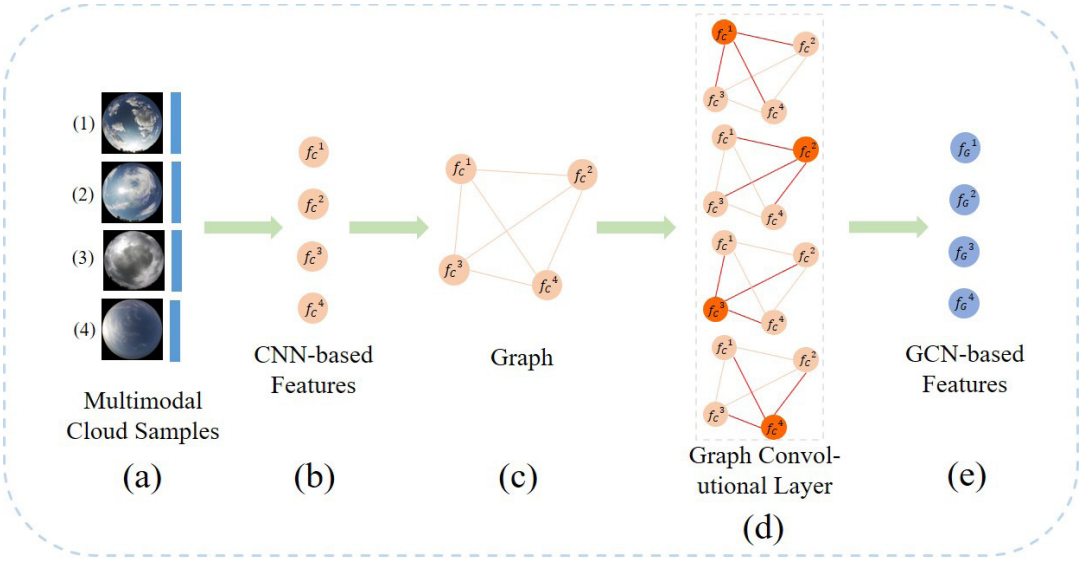


Fig. 4. Diagram of GCN working procedure. (a) Indicates the multimodal cloud samples. (b) CNN-based features of multimodal cloud samples. (c) Graph which is constructed based on CNN-based features. (d) Represents the graph convolutional layer. (e) GCN-based features of multimodal cloud samples. Best view when zoomed in.

where f_C^i and f_C^j are the CNN-based features of the ith and jth multimodal cloud samples in the graph. Based on the similarities, the adjacency matrix is formulated as

$$A = \begin{pmatrix} a_{11}, & a_{12}, & \cdots, & a_{1n} \\ a_{21}, & a_{22}, & \cdots, & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}, & a_{n2}, & \cdots, & a_{nn} \end{pmatrix} \tag{7}$$

where n indicates the number of nodes in the graph. When the graph has been constructed, the CNN-based features of multimodal cloud samples and the adjacency matrix are fed into the graph convolutional layer, and the operation of the graph convolutional layer is defined as

$$F_{G,l} = h(\widehat{A} F_{G,l-1} W^{l-1}) \tag{8}$$

$$\widehat{A} = \frac{A}{n} \tag{9}$$

where $\widehat{A}$ is the normalized adjacency matrix, W^{l-1} is the parameters to be trained in the $(l-1)$th graph convolutional layer, and $h(\cdot)$ indicates the ReLU activation function. $F_{G,l} = [f_{G,l}^1, f_{G,l}^2, \ldots, f_{G,l}^n]$ is the GCN-based feature of the $(l-1)$th graph convolutional layer. The input of the first graph convolutional layer is initialized by F_C, that is, $F_{G,0} = F_C = [f_C^1, f_C^2, \ldots, f_C^n]$. Specifically, $\widehat{A} F_{G,l-1}$ is used to integrate the information of neighboring nodes, and then the linear operation with W^{l-1} is conducted to update the GCN-based features.

To make GCN easy to understand, we take an example to show how GCN is working. As shown in Fig. 4, the graph is composed of four nodes, and GCN contains one graph convolutional layer. As shown in Fig. 4(b), CNN-based features (f_C^i) are first extracted from multimodal cloud samples using the CNN extractor, and then the CNN-based features are defined as nodes to construct the graph [see Fig. 4(c)]. We can see that the nodes in the graph are linked by the edges, in which the similarities computed by (10) indicate the correlations between nodes. Afterward, the graph is fed into the graph convolutional layer [see Fig. 4(d)]. In the graph convolutional layer, each node combines the information from the neighboring nodes to update its own features to obtain the GCN-based features. For example, the node f_C^1 could aggregate the information from its neighboring nodes f_C^2, f_C^3 and f_C^4 according to (8). Finally, the graph convolutional layer outputs GCN-based features of multimodal cloud samples as shown in Fig. 4(e).

D. Feature Fusion

In order to take full advantage of the two kinds of heterogeneous features and obtain discriminative representations for the classification task, we concatenate the GCN-based features and the CNN-based features to obtain the representations of multimodal cloud samples. The concatenator is defined as

$$F_A = \left[F_C^T, \mu {F_G}^T\right]^T \tag{10}$$

where $\mu \in [0, 1]$ is utilized to balance the importance of F_G. The concatenated feature F_A includes the visual information, the MI, and the related information, which could describe the multimodal cloud samples more accurately.

After concatenating the two kinds of heterogeneous features, F_A is fed to the classifier to predict which cloud type the multimodal cloud sample belongs to. We then follow the cross-entropy loss denoted as L_2. Hence, the total loss of the proposed HDF is defined as

$$L = L_1 + \gamma L_2 \tag{11}$$

where γ is a balance coefficient.

IV. Experiments

In this section, we evaluate the performance of the proposed HDF on the MGCD. We first introduce the MGCD and the implementation details of our experiments. Afterward, we

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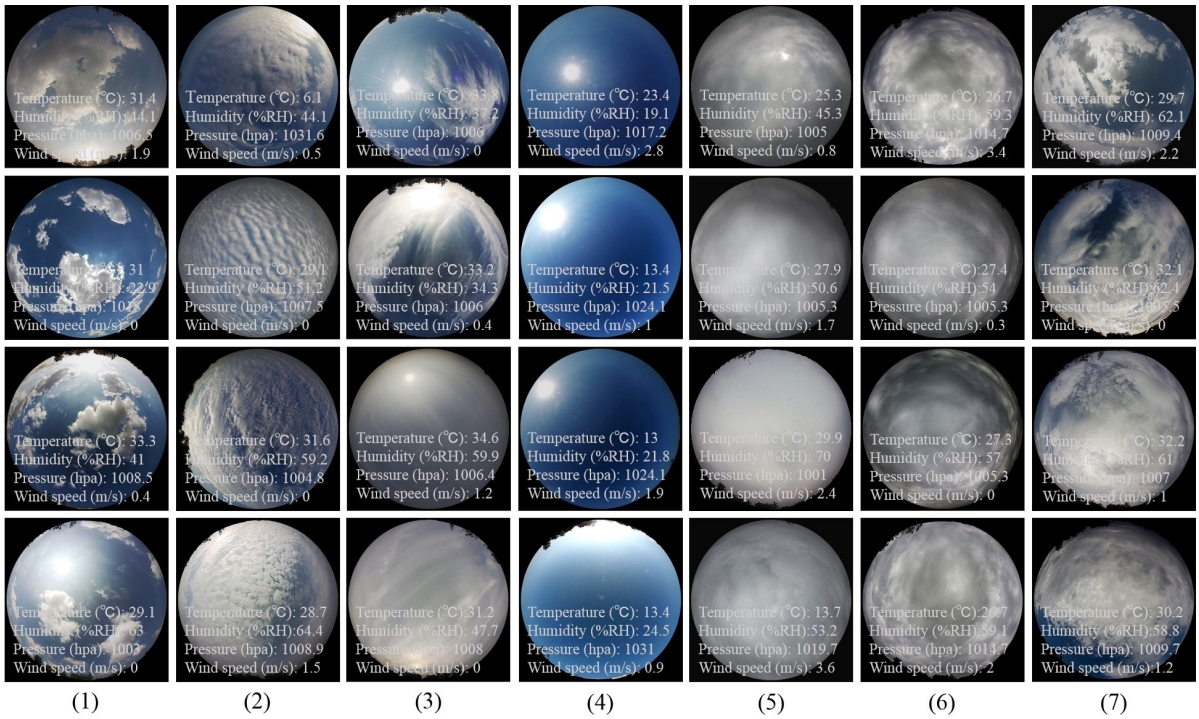


Fig. 5. Several multimodal cloud samples with seven types. (1) cumulus, (2) altocumulus and cirrocumulus, (3) cirrus and cirrostratus, (4) clear sky, (5) stratocumulus, stratus and altostratus, (6) cumulonimbus and nimbostratus, and (7) mixed cloud.

TABLE II
SAMPLE NUMBER OF EACH CLOUD TYPE ON THE MGCD

Cloud Type	Label	Sample Number
Cumulus	1	1,438
Altocumulus and Cirrocumulus	2	731
Cirrus and Cirrostratus	3	1,323
Clear Sky	4	1,338
Stratocumulus, Stratus and Altostratus	5	963
Cumulonimbus and Nimbostratus	6	1,187
Mixed	7	1,020
Total		8,000

show the experimental results to verify the superiority of the proposed HDF.

A. MGCD

The MGCD [50] is utilized to study the ground-based remote sensing observation, especially cloud classification. It consists of 8000 multimodal cloud samples with 7 cloud types, and Table II lists the sample number of each cloud type. Each multimodal cloud sample contains a cloud image and the MI. The cloud images are captured through a sky camera with fisheye lens, and therefore they possess a wide visual range. Furthermore, the MI is collected by a small weather station. The cloud images are stored in the JPEG format with the resolution of 1024×1024, and the MI is stored as a vector which contains four elements including temperature, humidity, pressure, and wind speed.

According to the cloud classification criterion published by World Meteorological Organization [51], the MGCD is divided into seven types, that is: 1) cumulus; 2) altocumulus and cirrocumulus; 3) cirrus and cirrostratus; 4) clear sky; 5) stratocumulus, stratus and altostratus; 6) cumulonimbus and nimbostratus; and 7) mixed cloud. Table III lists the descriptions of different cloud types on the MGCD. Several typical multimodal cloud samples with different types are shown in Fig. 5, where the MI is embedded in the cloud image. The MGCD is divided into training set and test set, and both of them include 4000 multimodal cloud samples.

B. Implementation Details and Evaluation Criterion

In the training stage, we foremost conduct the preprocessing and data augmentation strategies on multimodal cloud samples. Specifically, cloud images are resized to 256×256 and normalized by mean values and standard deviation values. The vertical and horizontal flips are conducted with the probability of 0.5, and the image rotation is implemented in the range of $[-25°, 25°]$. The MI is normalized to $[0, 255]$.

The weights of FC layers in the CNN extractor are initialized by a standard normal distribution and the biases are initialized by 0. As for the GCN extractor, we set the number of graph convolutional layer to 1, and the output channel of graph convolutional layer is 512. The total training epoch is 80. The learning rate is initially set to 1.5×10^{-4} and it decays at epoch 15 and epoch 35 with the decay rate of 0.2. The stochastic gradient descent (SGD) is employed to train the proposed HDF. In the test stage, the concatenated features generated by the concatenator are treated as the final representations of multimodal cloud samples.

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TABLE III
DESCRIPTIONS OF EACH CLOUD TYPE ON THE MGCD

Cloud Type	Descriptions
Cumulus	Puffy clouds with clearly defined edges, flat bottom and raised top, white or light-grey
Altocumulus and Cirrocumulus	Patched clouds of small cloudlets, mosaic-like, white or grey
Cirrus and Cirrostratus	Thin clouds, fibrous, white feathery clouds of ice crystals or sky covering, whitish
Clear Sky	No cloudiness or a very few cloudiness, blue
Stratocumulus, Stratus and Altostratus	Lumpy layer of clouds, broken to almost overcast, causes fog or fine precipitation, white or grey
Cumulonimbus and Nimbostratus	Dark, thick clouds, mostly overcast, indistinct outline, vigorous vertical development, thundercloud or rain cloud, grey
Mixed	More than one type of clouds

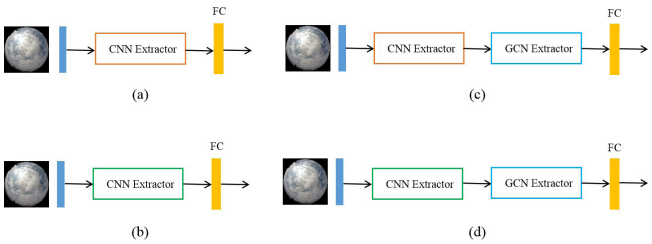


Fig. 6. Frameworks of the four ablation methods. The CNN extractor with an orange box is the vanilla version, while the CNN extractor with a green box is the full version. The frameworks of extracting (a) CNN-based feature of vanilla version, (b) CNN-based features of full version, (c) GCN-based feature of vanilla version, and (d) GCN-based feature of full version.

In our experiments, we utilize accuracy to evaluate the performance of the proposed method, and the accuracy is defined as

$$\text{Accuracy} = \frac{\sum_{j=1}^{J} m_j}{N} \tag{12}$$

where J is the number of cloud types, m_j is the number of correctly classified samples in the jth cloud type, and N indicates the total number of multimodal cloud samples in the test set.

C. Experimental Results

1) Ablation Studies: The proposed HDF integrates two kinds of heterogeneous features to represent multimodal cloud samples, that is, CNN-based features and GCN-based features. We perform the ablation studies to validate the advanced performance of different features in HDF including the CNN-based feature of vanilla version f_{CV}, the CNN-based feature of full version f_{CF}, the GCN-based feature of vanilla version f_{GV} and the GCN-based feature of full version f_{GF}.

a) CNN-based feature of vanilla version: The sketch of extracting the CNN-based feature of vanilla version f_{CV} is shown in Fig. 6(a). It utilizes the CNN extractor with the shallow fusion strategy to fuse the visual feature and the multimodal feature. Then f_{CV} is fed into a FC layer with seven neurons. Finally, the multimodal cloud sample is represented by f_{CV} with the dimensionality of 4096.

b) CNN-based feature of full version: As shown in Fig. 6(b), it is utilized to generate the CNN-based feature of full version f_{CF}. It fuses the visual feature and the multimodal feature using the deep fusion strategy. f_{CF} has the dimensionality of 6144.

c) GCN-based feature of vanilla version: In Fig. 6(c), the multimodal cloud samples are first fed into the CNN extractor of vanilla version, and then the GCN extractor is applied to generate the GCN-based feature of vanilla version f_{GV}. The dimensionality of f_{GV} is 512, and the neuron number of FC layer is 7.

d) GCN-based feature of full version: The sketch of extracting the GCN-based feature of full version f_{GF} is shown in Fig. 6(d). The multimodal cloud samples are first fed into the CNN extractor of full version, and then into the GCN extractor. The FC layer has seven neurons.

TABLE IV
COMPARISONS WITH DIFFERENT ABLATION METHODS

Method	Accuracy (%)
f_{CV}	86.30
f_{CF}	87.90
f_{GV}	87.80
f_{GF}	88.50
HDF_V	**89.60**
HDF_F	**90.08**

Table IV lists the classification results of different ablation methods. Herein, HDF_V indicates the proposed HDF employs the CNN extractor of vanilla version, and HDF_F represents the proposed HDF utilizes the CNN extractor of full version. From the table, several conclusions can be drawn. First, we can see that the proposed HDF including HDF_V and HDF_F achieves the best results. Second, f_{GV} and f_{GF} obtain higher classification results than f_{CV} and f_{CF}, respectively, verifying the effectiveness of the GCN extractor. It is because the GCN-based features consider the related information among multimodal cloud samples and enhance the discrimination of representations. Third, HDF_V and HDF_F achieve better performances than f_{GV} and f_{GF} because the concatenation operation executed on CNN-based features and GCN-based features fully integrates the visual information, the MI and the related information of multimodal cloud samples.

2) Comparisons With State-of-the-Art Methods: In this part, we compare HDF with state-of-the-art methods. The classification results are listed in Table V, in which there are two kinds of methods, that is, traditional methods and CNN-based methods. The traditional methods extract hand-crafted features of cloud images, and then directly concatenate MI with hand-crafted features for the representations of multimodal

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TABLE V
COMPARISONS WITH STATE-OF-THE-ART METHODS

Method	Accuracy (%)	Method	Accuracy (%)
BoW	66.15	BoW+MI	67.20
PBoW	66.13	PBoW+MI	67.15
$LBP_{(8,1)}$	45.38	$LBP_{(8,1)}$+MI	45.25
$LBP_{(16,2)}$	49.00	$LBP_{(16,2)}$+MI	47.25
$LBP_{(24,3)}$	50.20	$LBP_{(24,3)}$+MI	50.53
$CLBP_{(8,1)}$	65.10	$CLBP_{(8,1)}$+MI	65.40
$CLBP_{(16,2)}$	68.20	$CLBP_{(16,2)}$+MI	68.48
$CLBP_{(24,3)}$	69.18	$CLBP_{(24,3)}$+MI	69.68
VGG-f [57]	77.95	DMF [28]	79.05
ResNet-50 [21]	83.15	ResNet-50+MI [21]	84.48
Dev et al. [12]	67.84	JFCNN [27]	84.13
CloudNet [14]	79.93	HMF [29]	87.9
-	-	HDF_V	**89.60**
-	-	HDF_F	**90.08**

	1	2	3	4	5	6	7
1	0.99	0.00	0.00	0.00	0.00	0.00	0.01
2	0.01	0.82	0.00	0.00	0.12	0.00	0.05
3	0.00	0.00	0.91	0.01	0.00	0.00	0.08
4	0.00	0.00	0.00	1.00	0.00	0.00	0.00
5	0.00	0.01	0.00	0.00	0.92	0.07	0.00
6	0.00	0.00	0.00	0.00	0.11	0.89	0.00
7	0.07	0.21	0.04	0.00	0.00	0.05	0.63

Fig. 7. Confusion matrix of the proposed HDF_V.

	1	2	3	4	5	6	7
1	0.99	0.00	0.00	0.00	0.00	0.00	0.01
2	0.00	0.82	0.00	0.00	0.13	0.00	0.05
3	0.00	0.01	0.90	0.01	0.00	0.00	0.08
4	0.00	0.00	0.00	1.00	0.00	0.00	0.00
5	0.00	0.00	0.00	0.00	0.92	0.08	0.00
6	0.00	0.00	0.00	0.00	0.10	0.90	0.00
7	0.07	0.19	0.07	0.00	0.00	0.00	0.67

Fig. 8. Confusion matrix of the proposed HDF_F.

cloud samples. The hand-crafted features contain bag-of-words (BoW) [52], pyramid BoW (PBoW) [53], LBP [54], and completed LBP (CLBP) [55].

BoW learns cloud representations based on the frequency statistic of codewords which are obtained by K-means clustering algorithm on the SIFT features extracted from cloud images, and the number of codewords is set to 300. PBoW considers the spatial information of cloud images based on BoW. It divides the cloud image into 21 cells with three levels, and then concatenates the frequency statistic of each cell based on 300 codewords. Hence, the dimensionality of PBoW is 6300. LBP computes the binary value according to local region gray difference, and CLBP considers three factors including the sign and magnitude of local region gray difference, and the binary value of center pixel. Both LBP and CLBP have two important parameters, that is, the number of neighbors P and the radius of local region R. We conduct experiments on LBP and CLBP with different combinations of (P, R), that is, $(8, 1)$, $(16, 2)$, and $(24, 3)$, and both LBP and CLBP apply the rotation invariant uniform patterns.

The CNN-based methods consist of VGG-f, ResNet-50 and state-of-the-art methods. VGG-f [56] contains five convolutional layers (conv1-conv5) and three FC layers (fc6, fc7, fc8), and the output of fc7 is treated as the feature representation of cloud image. ResNet-50 [21] contains 16 building blocks, and the output of average pooling layer is treated as the representation of cloud image. ResNet-50+MI means the deep feature of cloud image is directly concatenated with the MI to represent the multimodal cloud sample.

From Table V, several conclusions can be drawn. First, our methods (HDF_V and HDF_F) obtain the classification results of 89.6% and 90.08% which are higher than state-of-the-art methods. Second, the performance is improved when using the MI for both traditional and CNN-based methods. It is because the MI contains different semantic information, which is beneficial to completed cloud representation. Third, HMF and JFCNN outperform the methods which directly concatenate the MI. It is because HMF and JFCNN optimize the visual and multimodal features in a unified deep learning framework. Finally, the classification results of HDF_V and HDF_F are 5.47% and 2.18% higher than JFCNN and HMF, respectively, because they apply GCN to provide the related information among multimodal cloud samples.

In order to further analyze the classification performance of HDF, we utilize the confusion matrix to show classification accuracies of each cloud type. Figs. 7 and 8 are the confusion matrices of HDF_V and HDF_F, where the numbers in each row indicate percentages of accurate classification for each type of clouds. We can see that HDF_V and HDF_F have the best results for the fourth cloud type (clear sky). However, they are weak in the seventh cloud type, because the seventh cloud type is the mixture of several cloud types and HDF is easily confused.

3) Parameters Analysis: In this section, we evaluate the influence of hyperparameters in the proposed HDF. The hyperparameters mainly contain the batch size in training stage, the number of graph convolutional layers, the coefficient μ in (10), and the coefficient γ in (11). We also evaluate the influence of the number of FC layers in MLP.

a) Batch size in training stage: The batch size is relevant to the construction of graph and the adjacency matrix of HDF. Hence, we conduct the experiment with different batch sizes, and the results are illustrated in Fig. 9. From the figure, we can see that both HDF_V and HDF_F obtain the best results when the batch size is equal to 16.

b) Number of graph convolutional layers: In order to show impact on the performance of HDF, we conduct the experiment with different graph convolutional layers. Fig. 10 illustrates the classification results of HDF_V and HDF_F where we obtain the best results when the graph convolutional layer number is set to 1.

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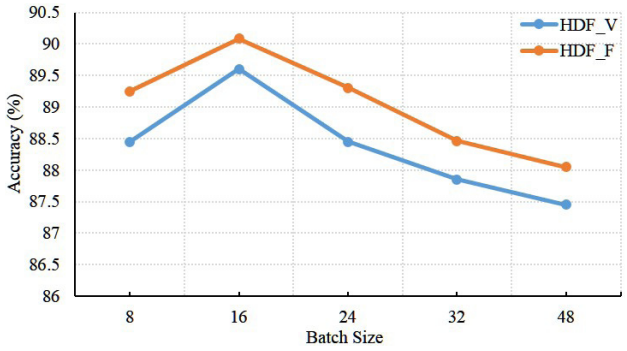


Fig. 9. Classification results of HDF with different batch sizes.

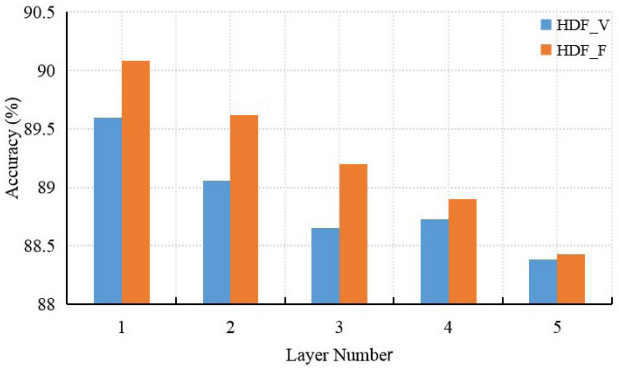


Fig. 10. Classification results of HDF with different number of graph convolutional layers.

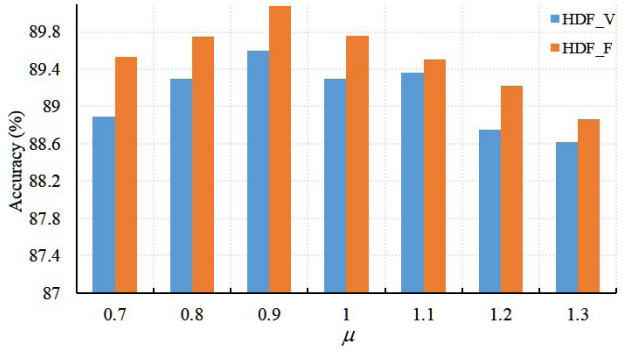


Fig. 11. Classification results of HDF with different μ.

c) Coefficient to fuse the two heterogeneous features: In order to take full advantage of the two heterogeneous features, the concatenator of HDF employs μ in (10) to balance the importance of CNN-based features and GCN-based features. The classification results with different μ are listed in Fig. 11, where we can see that the classification results increase when μ gets bigger while the results decrease after μ exceeds 0.9. Hence, we set $\mu = 0.9$ in our experiments.

d) Coefficient to compute the total loss value: In the total loss of HDF, γ is treated as a balance parameter, and the classification results with different γ are shown in Fig. 12. When γ is equal to 1, HDF_V and HDF_F both achieve the highest classification results.

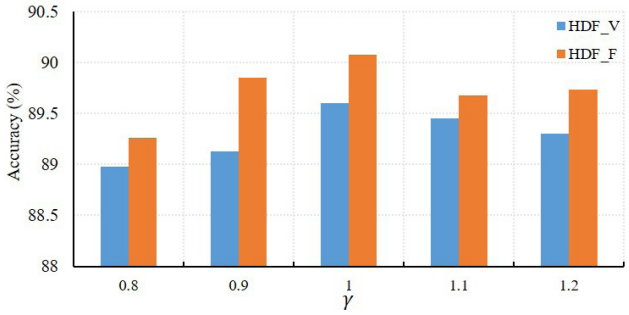


Fig. 12. Classification results of HDF with different γ.

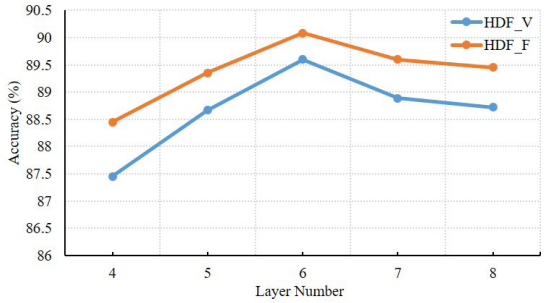


Fig. 13. Classification results of HDF with different number of FC layers in MLP.

e) Number of FC layers in MLP: In the proposed HDF, we utilize MLP to learn the multimodal feature. In order to test the influence of the number of FC layers in MLP, we conduct several experiments using different number of FC layers as shown in Fig. 13. From the figure, we can see that when the layer number is set to 6, both HDF_V and HDF_F obtain the best results.

V. Conclusion

In this article, we have proposed HDF to learn heterogeneous features of multimodal cloud samples, i.e., CNN-based features and GCN-based features. We apply two fusion strategies in the CNN extractor to obtain the CNN-based features. Then, the GCN extractor is utilized to generate the GCN-based features which could integrate the information of other multimodal cloud samples. In order to fully mine the information contained in the two heterogeneous features, we concatenate them to represent the multimodal cloud samples. To the best of our knowledge, we are the first one to apply GCN to the cloud classification task. The experiments performed on the MGCD demonstrate the effectiveness of HDF.

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